LifePulse360 Beneficiaries-Patient-Caregivers Empowering Technology: Response to Technology to Empower Medicare Beneficiaries

John Dodd from Dodd Health Innovation LLC with Snowflake input

Section 1: Summary Recommendations

Focus on Beneficiaries -Patient-Caregiver Engagement and Process Coordination with LifePulse360 User Interactions and specialized agent-technology aligned with the Snowflake Technology Framework

Request and Response to Request for AI-Based Agents and New Treatment and Care delivery capabilities can result in improved quality of life and reducing unnecessary and expensive hospitalizations and emergency room visits. Resulting in significant cost savings.

There is a better way for the beneficiaries for Medicare, Medicaid and Insurance Exchange to receive smart advice on understanding benefits capabilities and your specific needs.LifePulse360 with it's patient-caregiver focus is the better way forward for interventions that prevent and improve the treatment and care management. A multi-pronged simultaneous approach with a family of patient-caregiver eAgents build with new agentic AI technology. It can start with the use of eNavigators to guide and map your personal and precise needs to insurance programs in your community. It can support the creation of an Advanced Care Plan for the patient-caregivers, for family and for use with the treatment providers and care support teams. It guides in defining a personal blueprint for daily, weekly and life course management that adapts to ever changing needs. A personalized, engaging and empowering environment using patient-caregiver agentic ai technology is built for you and your caregiver. Agentic AI enables the creation using a solid foundation such as Snowflake that can be extended with emerging medical and personalized large language models. It is recommended that care coordination standards efforts be revived within ASTP/ONC. and the concepts and underlying coordination patterns and key information gathering, sharing, and reminding, alerting and recommendations be built with the solid agentic AI and large language models that have made a dramatic change over the last five years. This is an area that has not received enough attention but can both improve the life of the patients and caregivers but results in lower cost through preventable hospital admissions and improve treatment compliance and result in timely care interventions.

20th Century Personal Practices to Discover Rare Diseases and Get Control of Personal Information: A New Approach is Needed

John had years of Health IT integration and consulting for CMS, NIH, ONC and state Medicaid programs writing a book HealthCare Transformation getting the right treatment for his wife, caring for her at hospitals, home and in assisted living facilities made the challenges real and coping with the systems complexity and knowing the improvements were being thought about based on the "21st Century Cures" act John turned his frustration into an design with new AI-Based approach- LifePulse360. John has spent years listening to and living with the challenges and decided to cope his "purpose" was to leverage his knowledge and personal experience to create a new approach.

The United States has the best experts in medicine and rare diseases in the world but the systems and information gathering capabilities are missing. As John Dodd was faced with guiding the discovery of his wife of over fifty years the interoperability barriers mad it a great challenge. John created a personal health data management approach by gathering information from multiple emergency rooms, providers and hospitals to take to experts in another states. This all came to head at a rural hospital when stroke and suspected brain cancer resulted in move to another state and a large hospital with only the information that John had collected. Discovering what turned out to be two rare diseases took 9 and 12 months and actually over 5 years to get treatment and evolve a care plan. A technologist and problem solver like John new that transformation was possible.
While the 21st Century Cure Act was being discussed and refined in standards meeting and TEFCA/Common agreements. Getting information involved being the personal information gatherer and access information through multiple portals and creating adobe pdfs and passing them in messages to the experts to gather the gather the background information. Agentic AI technology can streamline this process and refine searches, such as asking, "Has she ever had a hematology consult?" "When was the EEG tests conducted and get us access?" "Gather all brain MRI's?" Then the health providers were sent with faxes and were then created into attachments. In some cases, the promised data was never delivered and the MRI/CAT scans need to be repeated. Additional costs.
These all could be formed into special eAgent commands as defined in this paper. We can move to the 21st Century approach and there can be a new approach. John has also lived in three different situations: home trying do most of the caregiving himself, assisted living facility with significant workforce changes, and moving to retirement apartment with home health aides. The ideas and experiences have been translated into LifePulse360 and some features such as an "Ounce of Prevention".

This focused RFI response addresses the needs define by the May 13, 2025 Announcement said: "The goal is to unlock the power of modern technology to help seniors, and their families take control of their health and well-being, manage chronic conditions, and access care more efficiently.

"We are building a future where seniors and families have the digital tools they need at their fingertips—tools that help them make informed choices, manage chronic conditions, and stay healthy," said Stephanie Carlton, CMS Chief of Staff and Deputy Administrator. "The government must be a catalyst, not a barrier, to unleashing American innovation in health care." **"**

Dr. Memhet OZ comments at May 21st conference said big changes are necessary and called out for help and ideas and our team is ready to share our ideas. In particular, Dr. Oz said

"I came to recruit you to help," Oz said. "The magnitude of the challenge that we're facing is scary, but you never want to waste a crisis. This is a generational opportunity for people to come and help play a role in whatever way possible."
The impetus for change can be seen in several aspects, he said, including the increasing strain of Medicaid on federal and state budgets, the much-publicized disparity between spending and population health outcomes, and the prevalence of medical errors.
"We need more sophisticated, multifactor approaches to many of the challenges that we've [discussed]," Oz said. "I actually think they're doable because the political will to make some of these seismic shifts exists now."

A series of recommendations are made along with optional backup detail is available on request to our team , if desired. Our team of Dodd Health Innovation consultants and the Snowflake technology team recommend the following.

2 Empowering Technology with LifePulse360: Beneficiaries-Patient-Caregiver eAgents

Summary Recommendation and Design elements

1. Understanding of Beneficiary Challenges with a series of use case scenarios and creating a Health Innovation Technology Readiness and Vision Document
2. Built Patient-Caregiver Centric US Core Data Items that can be the baseline for the Next Generation of AI/Data Models in the Cloud with the use of Protected Virtual Private Networks based on TEFCA
3. Create a Pilot/Test Sandbox with Leveraging LifePulse360: Patient-Caregiver Support (from Dodd Health Innovation LLC) and Snowflake.
4. Create a Multi-Blue-Button like interface with a Panel of Graphic Icons and links to underlying eAgents User Support Agents built on an agentic AI framework.
5. Create an "Ounce of Prevention" personalized prevention that involves a quarterly review of nurses and physical therapy specialist that recommends small prevention steps to keep older adults that improves the quality of life and prevents expensive hospital visits.
6. Test with a small sample of volunteers the prototype and research the implementation and deployment options. Review Medicare Advantage options and Value-Based Purchasing new models.
7. Define a governance structure for Empowering and Understanding of the Challenges along with the Navigation Issues of Systems to System communication that must be integrated information and sharing across distributed health and payer information sources based on series of information exchange scenarios
8. Create User Engagement and Empowerment Interface and Process Action Tools that leverage Snowflake Underlying Technology that can be used to build the smart information exchanges and process action lists that address beneficiary-patient-caregiver challenges
9. Provide a great environment where disease foundations and problem-solving advocates for change can use "empowering technology" to create a new trajectory for change.
10. Recommend ASTP/ONC and CMS Beneficiaries Standards with a Transformational Ecosystem with Personal Controls and Consent Management that can be defined with Open NIEM with a Health Services committee that can invite organizations such as the National Health Council and National Organization of Rare Diseases. The ASTP/ONC and CMS efforts should include:
    a. USCDI for Beneficiaries and Beneficiaries Personalized Trusted Exchange Framework: and a set of data and knowledge exchanges define with Open National Information Exchange NIEM) forming health-patient focused care committee
    b. for "beneficiaries to have the choice of downloading and Improve their Treatment and Care Management with their own control and a personal treatment and care tracking dashboard
    c. Define a Personal Health Data Trusted Exchange Framework- that will be built upon TEFCA
    d. Use eAgent Integration Agents to build a History of Encounter Data using agentic ai
    e. Create a Health Event History and prevention care plan
    f. Create a prevention Brain Health and personal knowledge model using a personal knowledge gathering agent it will collect both knowledge on specific conditions from rare disease foundations and from groups such as the Cochrane foundation
    g. A personal stress and agitation assessment and daily management process will provide a personal advise based guide for the caregivers. It will be built upon psychological practices on "behaviors that challenge" process-advise personalized response approaches. Agitation effects about 30% of dementia patients but techniques to manage it have shown great positivity but delivery has been difficult.
    h. Create a Medicare-Benefits Plan and choose the Medicare Advantage Options that fit your situation

3 Empowering Technology with LifePulse360: Beneficiaries-Patient-Caregiver eAgents

i. Encourage high risk patients an Advanced Care Planning to reduce the acute care needs of high-risk patients and those with disabilities and link to professional care coordinators and home health agencies
j. Define Value-Based Metrics and Dashboards for Personal and Population Health Tracking
k. Use of eAgents and Graphical and Data Linked Models and link to agentic AI and specific medical large language models
l. Define Knowledge Exchanges with Open NIEM and define channels of communication for request and reply, event driven data and knowledge exchanges along with subscription and event delivery alerts

11. Reducing the Burden of Compliance with New Program Integrity Checks and Giving the Beneficiary Choices within the Medicare Advantage Programs and will streamline Medicaid Enrollment as well.

Our approach to define treatment and care management interventions that reduce the need for acute care and improve the quality of life for beneficiaries can result in personal, population and Improvement in Medicare Delivery of Services.

Change toward a Patient-Caregiver approach will need the consultants an advocates such as John Dodd to reach out to innovative AI-platforms like Snowflake and form collaborative partnership. The RFI process has opened the dialog and now we are eager to discuss, demonstrate our ideas and approach, and undertake development sprints in collaboration with ASTP/ONC and CMS to create a beneficiary empowerment ecosystem.

Collaborative Partners are being sought to refine the concept and to support the data and knowledge extraction and personalization process Currently we are in conversations with National Organization of Rare Diseases (NORD), University of Maryland Center for Aging Research, University of Delaware Nursing School and are interested of being connected with others to continue the concept validation and refinement process. Work with health care organizations such as Assisting Hands and the Association of Prevention Medicine to create and refine the concepts for "Ounce of Prevention" and preventive brain health and agitation management based on Ian Andrew James work on "Behavior that Challenge" . These collaborative efforts will result in an eBook, demo and investment plan and a refined open-collaborative governance process .

Please contact John Dodd -410-598-7266 john.dodd@bdc-healthit.com

LifePulse360 Beneficiaries-Patient-Caregivers Empowering Technology: Response to Technology to Empower Medicare Beneficiaries

John Dodd from Dodd Health Innovation LLC with Snowflake Technology Material

Section 2: Detailed Recommendations Report

Introduction

Current Health IT Capabilities have been focused on the Hospital and Providers office-user levels, with little consideration given to patients and caregiver needs. Medicare benefits are complex and confusing and the needs very greatly. A vision for a new approach has been defined by a startup who has been working to define a care coordination platform that turns all information to be patient-caregiver centric. This has been both a personal challenge faced by a problem-solver with extensive system of systems experience and mired in the complexity of caregiving world where "21st Century Cures" and the ASTP/ONC ideas have not been implemented and the power of AI is ready to enable a new way of treatment and care delivery. The results is the early stage LifePulse360 and the detailed readiness assessment of AI for Medicine and the new platforms that can be a foundation for empowering care delivery reform. In response to the RFI and the encouraging words from the CMS and ASTP/ONC leadership, we like to share a vision for an empowering ecosystem that can evolve offering Medicare and Medicaid innovations that can spark care delivery reform that improves complexity management, fosters prevention and early intervention that results in less costly care and fewer emergency room visits and preventable hospitalizations.

LifePulse360 coordinates care for high-risk patients and those with family or professional caregivers. With a cursory 20-minute annual wellness check and poor understanding of what benefits are available beneficiaries start each annual enrollment without a real care plan. Understanding their insurance options from a vast array of Medicare Advantage plans and supplements overwhelm too many Medicare patients. Patients don't have a treatment and care management plan and wait for an acute incident to drive any life changes. Health care can be uncoordinated and sometimes overlooked, leading to unnecessary costs. The LifePulse360 platform design and pilot implementation was based on both knowledge and consulting with HHS and state Medicaid programs and extensive experience in using early AI in the space program and counter-terrorism integration but tracking AI that was a hidden gem. As TEFCA and USCDI evolved AI techniques came out of the shadows and Health Innovation became possible. AI technology with Safety Features have recently been revealed. New AI-Cloud-based a stable foundation by companies like Snowflake are ready for deployment and can handle the scale and phased improvements over the next 5-10 years. Like many of the CMS/ASTP/ONC leaders John Dodd knew through personal exposure and personal involvement with beneficiary-patient-caregiving needs the g challenges faced with a lack of an integrated system of systems. A consultant facing these challenges has used his problem-solving mindset and tracking of CMS-ONC -HHS knowledge progress to create a Target Vision for a new Ecosystem. This RFI response focuses on the neglected beneficiary-patient-caregiver tools and proposes an evolutionary ecosystem through extensions of existing standards and new distributed AI process and knowledge exchange infrastructure that can build upon foundations such as Snowflake.

Empowerment: Beneficiaries eNavigators with Interactive and Knowledge-based Shared Decision Making for Benefits Needs and Integrated Treatment and Care Plans

Far to often the Beneficiaries do not understand Medicare and don't know who to go to ask questions. A Medicare advantage salesperson talks up their plan and the beneficiary switches plan often year by year. After an emergency room visit or an acute hospitalization there are surprise bills or a wake-up call to change your behavior. But follow-up and true patient engagement defaults to being told to go to a portal for information. This does not have to be this way Health Innovation Technologies can create a New Way Forward.

There are calls to CMS 800 but the answers are not tied to a plan for either treatment or care plans. Putting information and decision-making power in the hands of the beneficiary was enabled by the "21st Century Cures Act". Through the work of ASTP/ONC standards and policy efforts but only an innovative company like Dodd Health Innovation LLC would take up turning health innovation into empowering and problem-solving for the patients and especially the high risk and high-cost patients.

High risk patients and those with rare diseases often see many doctors and hospitals, all with a part of the information puzzle. Poor discovery and treatment plans result when decisions are made by doctors who often do not have a complete picture of the medical issues. Then incomplete care needs of the patient and/or their caregivers' situation result in repeatable and preventable hospital visits. The "systems elements" are just not working for the beneficiaries, and it leads to frustration and complains. A more effective approach like eNavigators for planning and reminders, alerts and recommenders in your personalized daily schedule. Some of those examples are here but the empowerment tools allow grassroots development of personal patterns of treatment and care needs can be defined.

Create a CMS 2030 Interoperability Vision and New Update: Personal TEFCA, USCDI and Open eAgent Technology

The concepts of a care coordinator and caregivers with patients was developed by the LifePulse360 design team. In parallel CMS-ONC directions on interoperability were being generated. Some states had their own health innovation efforts such as the Delaware Health Commission. There have been efforts around with Federally Qualified Health Center and community accountability such as the City of Dover. LifePulse360 was working with a number of ideas for LifePulse360 were stopped and were impact by family caregiving responsibilities.

LifePulse360 was restarted in early 2025 with a focus on patient-caregiver challenges. The RFI has allow the effort to leverage some of the early ideas defined by the CMS Federal Register write-up and the directions of ONC. During 2024 and early 2025 a restart of LifePulse360 and use scenarios and readiness of AI for Health Care Delivery reform has included an alignment with Snowflake. We have been reaching out to other collaborative partners such as the National Organization of Rare Diseases (NORD) and having been a consulting to both ONC and CMS the review of the standards efforts and next steps are recommended. The preliminary recommendations are:

- current state of standards and CMS-ONC collaboration
- ONC has addresses interoperability issues and TEFCA, USCDI have been refined
- But patient-caregiver perspective and beliefs and key points from the federal registry have been defined and recommend a more detailed review
- Need to create an approach with the beneficiary as the central focus

- Summary of initial review and key points
- Medicare TEFCA- Payer TEFCA and Personal TEFCA and Provide USCDI and open shared services that can be part of empowerment governance community that can create aligned Profiles (Basic, Intermediary and Advanced)
- Use a set of Health Innovation Technology with eAgents that can Augment and Navigate the issues for the different perspectives- create a blue-button like panel of large icons and links to eAgents and user support agents
- Data Sharing Services that support "best available data services" and identify short comings that can be addressed with shared improvement approach and continuously improve
- Create a "we can improve environment with Empowering Tools and Open Governance including the use of Open NIEM and the creation of open shared services for data and knowledge exchanges that can support a range of innovative interventions and care delivery redesign e

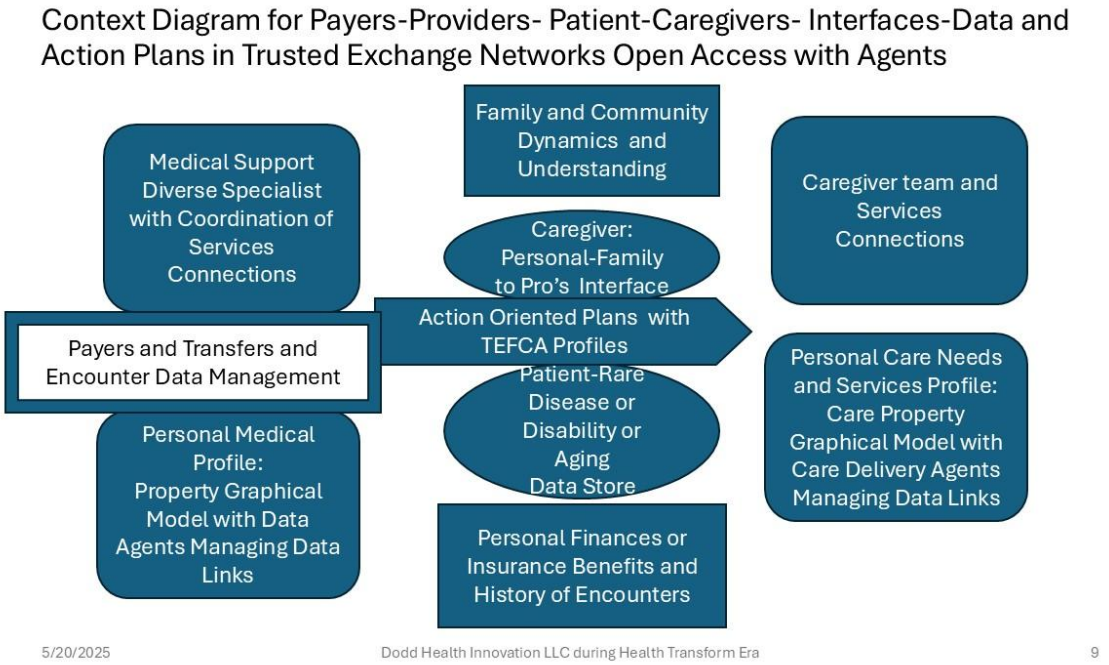
Further analysis in these elements and refinement of the LifePulse360 design and eAgent elements will be needed.

LifePulse360 Platform Solution with a Vision and Use of Empowering Technology

LifePulse360 is a platform for patient-caregiver information integration, shared decision-making, action tracking and advanced planning. It espouses a high quality of life, and addresses many challenging situations faced in life. AI and Digital Health Technology has not sufficiently addressed the patient-caregiver needs. LifePulse360 addresses them, with a highly personalized approach.

At present, patients' decisions to take preventive actions, or not, are not effectively managed. Caregivers, at home or in assisted living facilities, often struggle with integrating recommendations of multiple providers with whom they may actually disagree. An aligned care plan is needed; this can prevent situations from getting worse. An aligned care plan improves diagnoses and tracks symptom changes, especially for patients with rare diseases, who often see multiple specialists and take multiple medications, with no complete, accurate diagnosis, treatment and/or care plan. A series of information exchange scenarios can be defined.

The context diagram is shown below.



From Fragments of Beneficiary-Patient-Caregiver Information to an Integrated and Linked Data Model

Under the 21st Century Cure Act of 2016, the patients own and can get access to their health records. However, an implementation strategy and efficient approach has not been designed and implemented. LifePulse360: Patient-Caregivers Support defines a central property-based graphic model and data links, supported by data agents. These support agents translate health information into a commonsense format, understandable to the patient and caregivers. Lack of understandability is one of the reasons for non-compliance. Scheduling, and being reminded of schedules, are essential for being compliant with treatment. Treatment and caregiving are often a burden that the patient and caregivers can't handle and finding support is often difficult. The problems for the aging population and their specific diseases drive up costs, including choosing an expensive option, i.e., going to emergency rooms, even ending up in hospitals, often for repeat visits. Improving the delivery of care on a day-to-day and hour-by-hour basis can result in improved quality of life and save money.

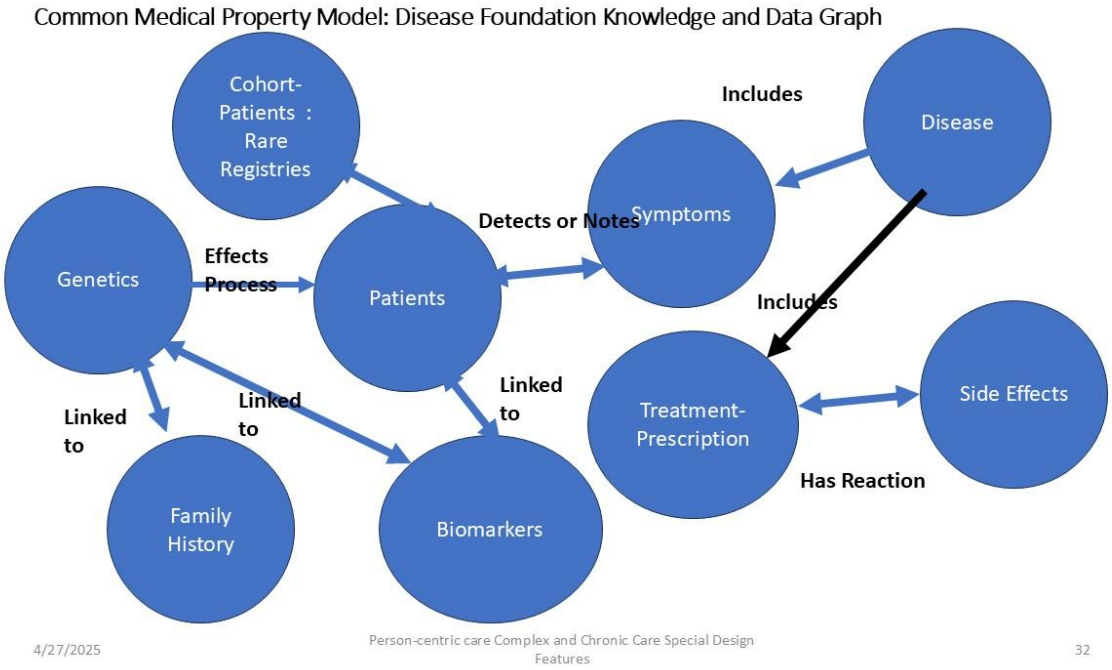
Today, most doctors and hospitals have data, test results, visit and consult notes and images and genetic and blood tests but they are stored in scattered databases. Since 2016, the Office of National Coordinator of Health IT and various standards bodies have slowly moved to create standards. It appears, however, that the only ones that care about data integration are the patients and their caregivers. The patients with one or more rare, hard-to-manage, or chronic conditions must struggle to manage the treatment and care plans. LifePulse360 transfers the power of health innovation technology to patients and their caregivers, creating for them the capacity to address their struggles firsthand. Moreover, LifePulse360 has been invented by a technologist who has experienced the struggles from a personal standpoint, over an extended period, and concluded "there can be a better way" to do this. This provisional patent application, employing a portfolio of smart eAgents, has evolved over the last few years of design and observation of patient-caregiver struggles.

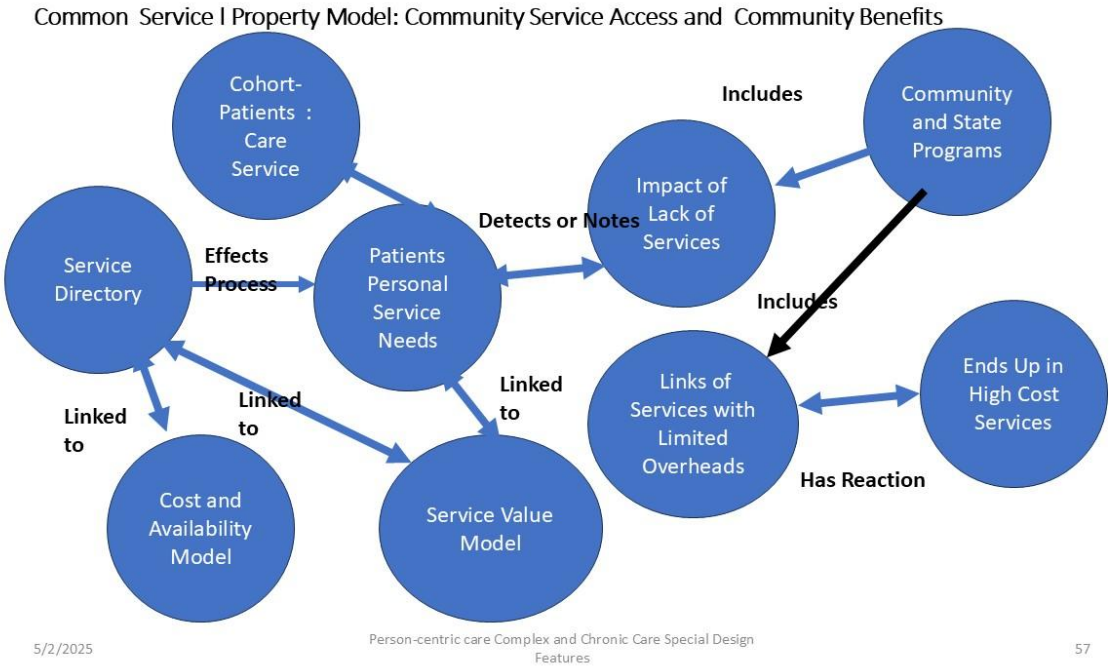
8 Empowering Technology with LifePulse360: Beneficiaries-Patient-Caregiver eAgents

Personalized Data Extraction is supported by Data Gathering agents

The data is distributed but can be gathering and linked to a common core structures that are called graphical data models. The first context diagram, shown below, reflects the patient and caregiver in the center, for both their treatment and care plans. The second and third diagrams define structured, graphical data models, with data links to other structured information sources, augmented by unstructured data sources. Each Beneficiary can have the option of gathering their data from payers and providers using the information gathering agents. The display and summary agents can provide a visual model of encounter data and progress data. This can help manage the diagnostic and treatment planning process.

The data models can be represented in Snowflake, agentic AI tools leveraging Snowflakes with focused patient-caregiver agentic AI design and empowering open sharing capabilities. The specific patient-caregiver agent design can be built with agentic AI capabilities of Snowflake. The USCDI elements can be used as core elements and be extended with beneficiary-patient-caregiver data elements and a defined set of datalinks that can be managed with data sharing agents. The initial graphic property data models are shown below.





For several years, there have been uses of such models utilized in scientific research, criminal investigations and distributed applications such as military, counterterrorism and integrated intelligence efforts. However, medicine and health care have been slow and hesitant adopters of innovation. As LifePulse360 patient-caregiver ideas evolved, agentic AI has come out of the shadows and new, leading edge companies like Snowflake have emerged.

Personalized Knowledge Gathering and Links to Trusted Knowledge Sources run by Disease Foundations

We are in the knowledge era and non-profit associations built by advocates and donors to the causes of rare disease have created registries of patients such as those sponsored by the National Organization of Rare Diseases (NORD) where there is an identification of patients that can be case studies and part of clinical trials. Researchers, providers and patients attend both in person and virtual annual conferences. Now is the time for the next step to take the "web resources, events, and collective knowledge from disease foundations' along with other places like NIH National Library of Medicine (NLM) and create a Learning Health Systems using Large Language Model technology. The many rare disease foundations could work together to build a personalized learning environment and active process environment.

As the disease information is refined with trials and evidence-based studies, levels of trust and maturity can be categorized. New User support agents can help define the additional categories of pertinent information becomes apparent, involving capturing the concepts, beliefs and preferences associated with the phase of life in which you are interested. The primary focus has been on the aging and disabled populations based on NIH Institute of Aging. Data pertaining to retirement practices and geriatric psychology is captured in a series of questions and interviews that can be used by the caregivers and families, addressing the inevitable

questions such as "what mom or dad would want to happen". This personal knowledge base can help guide end-of-life and decision-making with respect and dignity.

To provide compliancy and understanding of my wives two rare diseases and relate to the experts neurologist I have created a knowledge map and linked to the Foundation for Peripheral Neuropathy (FPN) and the Charcot-Marie-Tooth (CMT) Association along with Cerebral Amyloid Angiopathy (CAA) Forum and have create a personal knowledge library and created a summary of practical information. The next step is to create a pilot personalized large language model and link to a graphic knowledge mapping tool and determine how this can be scaled. Plan to present the ideas and discuss at the National Organization of Rare Diseases (NORD) fall 2025 Summit.

The applicability of domain-specific and specialized usages of agentic AI found in LifePulse360's eAgent Technology will be summarized under appropriate categories below.

Empowering Technologies Evolutionary Design based on Case Studies and Personal Insights

While reviewing the needs identified in case studies, a series of detailed eAgent Technologies has evolved. The agents have been designed to reduce the burden of treatment and care. These agents also address the complexity of care management along with the uncertainty and dynamics of the disease progression. The agents deliver information to the right person at the right time. The agents track the side effects of medication to track whether improvements are being made or are worsening. The health literacy of patients and caregivers converts many sources of information into understanding. Agent technology is not trendy. There has been a slow flow of advancement as AI gained acceptance. AI noticed agents again in 2022, but the Big Electronic Health Information vendors have not looked for innovation in healthcare. The health IT buyers have been administrators, doctors and hospitals with government subsidiaries. Very few vendors have looked at the difficulties of patients and caregivers like the author has. This unique understanding, developed over many years searching for innovative elements to help the patient and caregivers has evolved into the agent techniques described here. The work with NIH in research and Medicaid involves developing systems for care needs and sharing innovations broadly. As a result, eAgent portfolios have evolved into portfolios that can fill the "innovation gaps" and offer a more efficient, effective care programs and improve the quality of life for the patients and caregivers.

Extensions to Cloud-Based Property Linked Graphical Data Model for Patient-Caregivers

The concern expressed by the health systems was their need to identify as many data sources as possible that deliver the information in a timely fashion, prior to the patients complaining to the Office of Civil Rights at HHS, creating a perception that the organizations were not compliant with the law. Big hospitals tried to comply and fight against the regulations while a few disease foundations were advocating for patients' integrated infrastructures. ONC HIT and standards groups created pieces of a solution, but nothing has addressed the complete answer from the patient-caregiver point of view. Pieces of Artificial Intelligence called agents, and their process integration, have been designed to address these issues and to solve a set of special issues. The pieces of the puzzle have existed, but the tools the eAgents present, and process-progress tracking needed by patient-caregivers, were not realized and/or understood, until the inventor experienced a set of challenges. Some recommendations made possible by successful use of eAgents are:

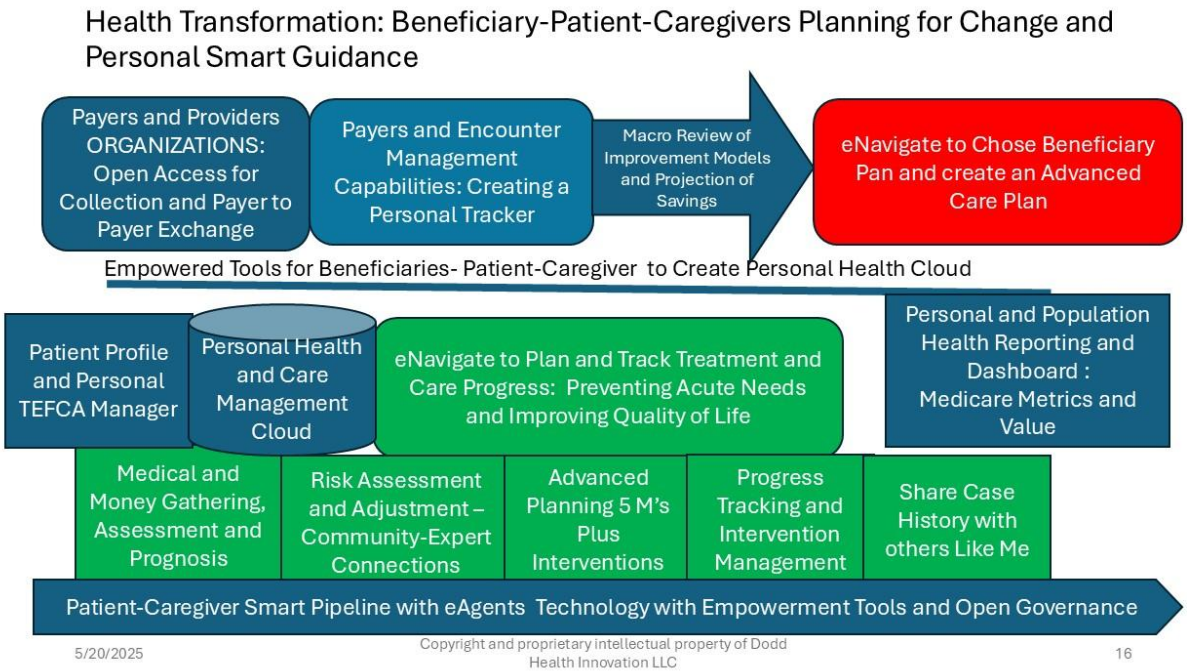
- Creating an Integrated Health Record for a Patient, based on Data from provider visits and hospital and emergency room care records.

- Transfer of information to/from patient portals, getting access to imaging and blood work, and sharing with a new specialist.
- Creating an integrated Treatment and Care Plan that fits a daily and periodic schedule of progress tracking and integrated interventions.
- Explaining the reason for the treatment with Health Literacy and Common-Sense Reasoning in a way that makes sense to the patient-caregivers.
- Managing the Data Consents, Trusts and taking Prevention-Protection Steps.
- Providing an adjustable and adaptable environment that addresses needs through telehealth and detailed assessments by experts; and
- Establishing a Value-Based and Cost-effective system of care management.

Empowering Capabilities are an Integrated Addon and Not a Replacement

The interface on the beneficiaries'

smart phone or computer interface that is connected to the many existing elements such as shown below. One example of how these technologies can support health transformation as shown below.



Follow Emerging Standards and allow for extensions and adjustments to data structures, labels and links

The patient-caregiver personal data-in-the-cloud can follow the United States Core Data for Interoperability (USCDI) Items can be selected for the Patient-Caregiver Core Data Elements. Additional care coordination elements and specific care management extensions that depend on the disease burden and situation faced by the patient (eg living alone, as a couple at home or in assisted living and a range of caregiver situations). The core care be linked to related connections such as to diseases registries and genetics registries or by the

type of treatment and care needs templates, The templates are adjusted and used for LifePulse360 for progress tracking. Using small intelligent agents can lighten the load of managing data and tracking treatment, care, and progress.

Next Stage of Assistant Secretary of Technology Policy Office of National Health IT

ASTP/ONC has created a data model USCDI that can be adapted and TEFCA can be adjusted to define the data exchanges between all the beneficiary data exchanges with a model such as the National Information Exchange Model(NIEM) that was initially used on counter- terrorism data exchanges but is now an open standard maintained by OASIS and can be extended with data service agents such as we propose. An Open Medicare Data Exchange Working Group can support the coordination and empowerment of Medicare key players and can include beneficiary representatives. One of the outcomes of such a group can become a standards and infrastructure element that the Center for Medicare and Medicaid Services (CMS) can provide in the form of registries, data stores and empowerment tools. The tools can include Profiles of Beneficiaries Standards with a Transformational Ecosystem with Personal Controls and Consent Management and previous HHS care coordination efforts and value-based coordination models should be revived.

The next generation of technology is now ready to build upon the work of ASTP/ONC and tailor it "empowering technology for beneficiaries". The Dodd Health Innovation and Snowflake teams can define a new Beneficiary- Patient-Caregiver core and linked data model with build on the USCDI and align with a graphic data model with data links. It will be for Beneficiary specific data elements and knowledge chunks that can be used for navigation through the constantly evolving Medicare rules. It will be designed to support the eNavigator functions and allow for a directory of benefits options. Building in smart navigation tools will greatly enhance the understanding of benefits and make the rules more explainable. Answers to common questions can be built in and all new federal registry rules can be designed as explainable knowledge elements as opposed to often unreadable and massive word descriptions.

It is recommended to create Beneficiaries Profiles for Trust and Consent management, tailored to Medicare needs, within the Trusted Exchange Framework Common Agreement.

A series of use case beneficiaries' scenarios should be developed, and a set of health innovation strategies should be developed and reviewed with a ONC/CMS working group.

A pilot implementation model can be built with Snowflake and "sandbox" created for testing and quick feedback and discussions at conferences. Implementation alternatives can be provided such as for "beneficiaries to have the choice of downloading and Improve their Treatment and Care Management with their own control management choices. The pilot project can consider the following:

- Define a Personal Health Data Trusted Exchange Framework-
- Use eAgent Integration Agents to build a History of Encounter Data
- Create a Health Event History and Personal LLM set of services
- Create a Medicare-Benefits Plan and choose the Medicare Advantage Options that fit your situation
- Encourage high risk patients an Advanced Care Planning to reduce the acute care needs of high-risk patients and those with disabilities and link to professional care coordinators and home health agencies
- Define Value-Based Metrics and Dashboards for Personal and Population Health Tracking
- Use of eAgents and Graphical and Data Linked Models
- Define a series of design and verification process to assure that safety, trust and evolvability are built into the design and implementation

Create a Library of Patient-Caregiver specific eAgents for easy Process Integration

The LifePulse360 design team has emphasized the creation of an intake and creation process for an Advanced Care Plan and tactical schedule link to process action lists and integrated with special purpose agentic software called eAgents that will be built with the Snowflake platform. Agentic ai platforms and underlying approach can enable the simultaneous focus on the patient-caregivers multiple concerns and allows concurrent process management to address complex treatment and care needs.

The patient-caregiver specific agents are summarized below.

The eAgent Technology focuses on seven specific areas:

- Use a Personal Information Gathering tool to collect data from various sources, identifying concerns such as genetics, health risks, mobility issues, mental conditions, available funds, and interactions of multiple factors. A set of care coordination actions can involve clarifications on medications, review of signs, symptoms and side-effects and referrals to others based on a concern is not resolved or understood and links to experts.
- Advanced Planning for Care Needs Agents can use the information to create 5 levels of needs from low to very high levels and identify risk adjustments that can be made.
- Personal Progress Tracking and Advisory Agents can take the beliefs and knowledge from the Advanced Planning Process and periodic adjustments to manage both daily and scheduled actions for the caregiver and links to the Medical Support Team and links to Care Supporters. The Personal Agents can remind you, alert and recommend and track the path you are on and the dangers ahead.
- Link, Trust and Consent Management Agents can link to portals, to trust information about you disease and connect to patient-registries and keep you up to data on the latest data about your conditions and connect you to experts. It manages the “wild west” of medical and healthcare information and includes protection agents that tracks the latest scams.
- Agents for information sharing and case reporting prevent unauthorized access to your data. Your data is yours and not the governments or a provider and you can share when you want. Population Health Information Sharing is made easy when they ask and you agree.
- Smart Call and Response Agent to address Potentially Preventable Situation faced by Senior Patients and Caregivers at Home and in senior assisted living situations by going from a dumb call button to a smart-timely response that reacts to needs of the situation
- Agitation and Stress Management agents that understand the type of dementia, personality, conditions that trigger agitation and include recommendation on how to handle “behaviors that challenge” that are so difficult for caregivers to handle
- Trust Tracking and Management with Knowledge-Validation and Refinement during adjustment, adaption and as part of continuous trust management process cycle.

These specific eAgents can be built and adapted with the Snowflake agentic AI tools and support a agentic ai collaborative open-source community of participants.

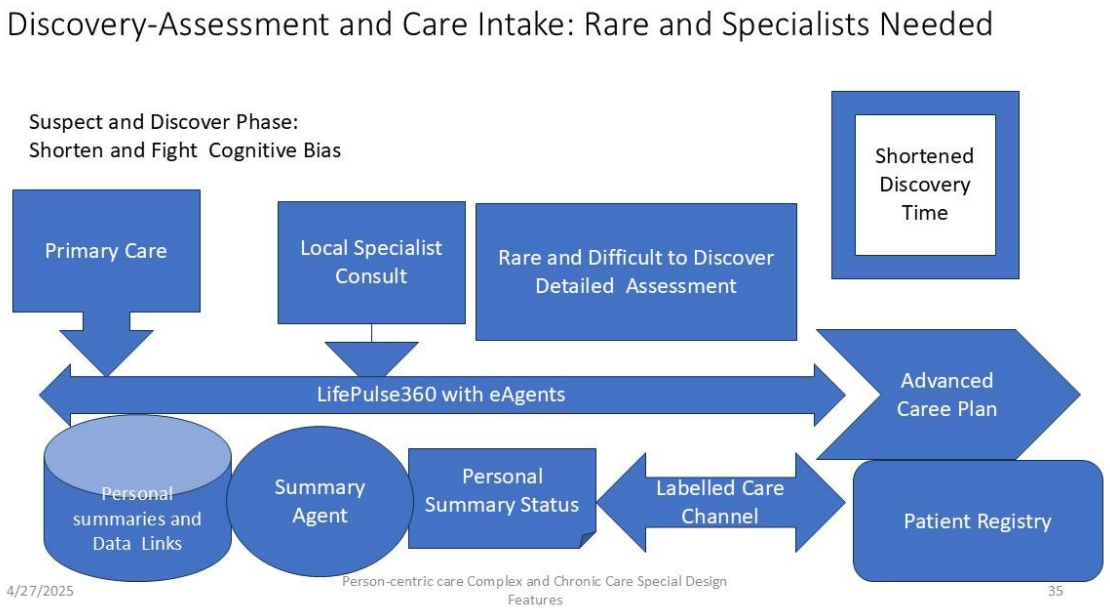
Personal Information Gathering Tools

The key focus of the patient-caregiver approach is to create an integrated view of the high risk and high utilizer of treatment and care needs. Those with rare diseases or chronic conditions often see many specialists and have multiple symptoms and side effects that results in many medical conditions,

medications and needs for care. But today the information is scattered and isolated. Over a 5 year period as patient Chris D was wandering through a series of emergency rooms, hospital stays and seeing specialists but not experts. The visit summaries, visit expert consultants, blood tests, images and the research in articles and research summaries collected in PowerPoints as the coordinator was trying to discover the conditions. It was known that health data standards could help but the "21st Century Cures Act" helped define standards and advocated the use of AI the vision of personal information gathering agentic tools has evolved. During the first quarter of 2025 using portals from the three different hospitals a series of visits, tests, results were gathered and then batched and shared with three new specialists that finally diagnosed the rare conditions Chris D was suffering and treatment/care plans were developed. Having used a previous generation of agent development technology, a set of designs for information gathering was defined to use Snowflake base tools with extensions defined with a high level summary shown below. A series of detailed refinements using these tools will be made available as a Snowflake partner. Specialist will guide the collection based on key terms and "find if patient x" had a consult with expert y during hospitalization or was an image of A take and how recently. Providers often ask caregiver or patient questions, and those questions should depend on the information gathered or create a logical request for structured or unstructured data or actually recommend two professionals talking to each other and a referral to a expert to rule out or in one of the many rare conditions.

The information gathered may also have unstructured questions of medical histories, past surgeries, accidents, family history and what specialists have you seen. Specialists have a great understanding of many rare conditions, and a set of rare disease discover questions – have you thought about this?
Even specialists may not often see a condition and need to see the patient's big picture and the symptom progression.

A series of special gatherings and discovery as possible diagnosis will fit as the detailed elements below the top-level flow shown below.



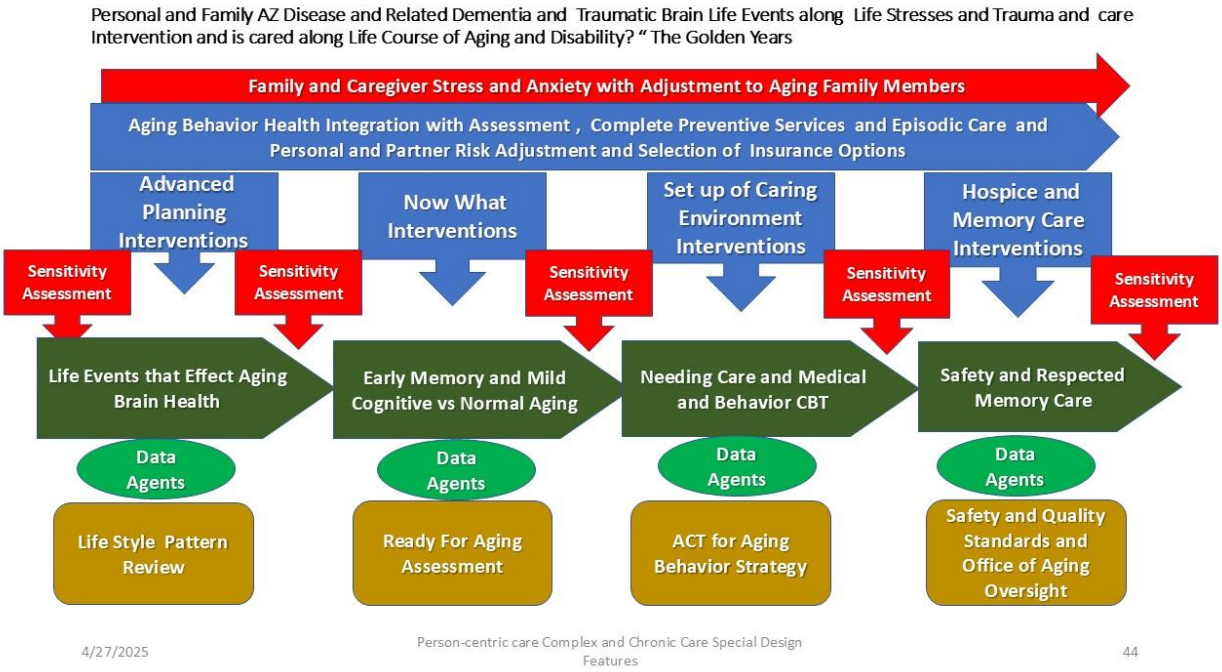
15 Empowering Technology with LifePulse360: Beneficiaries-Patient-Caregiver eAgents

Beneficiary Life Care and Advanced Care Planning and Adjustment to Changing Needs

Some are born with a disabling condition, and others are faced with accidents, crime, terror or behavior conditions or genetics that effect later in our lives. We have taken a life course approach and instead of waiting for the disease and genetic disposal to overwhelm us we want to have an advanced care plan for the well and the sick or future sick. Today, many but not all babies get wellness checks and folks on Medicare get annual visits where those hope to fake it through the simple cognitive decline questions and avoid discussing their obesity or smoking or lack of exercise. An approach to creating a life course approach and addressing the sensitive issue that follows a life period phase.

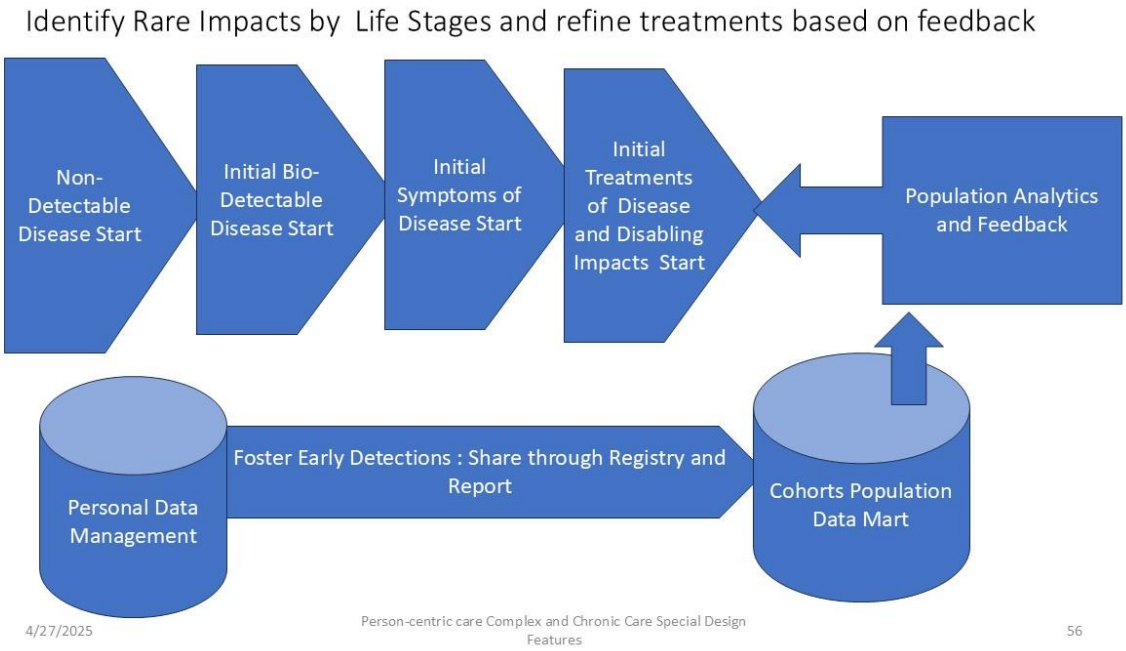
- Genetics can be detrimental factor but does not have to be a destiny. When individuals move into an assisted living facility or a life care multi-level facility, unstructured questions often arise. These should be systematically organized into an Advanced Care Plan, which should be periodically reviewed and adjusted as circumstances change.

The diagram below shows the life course representation.



Each rare disease and many chronic diseases have a few possible pathways to improvements, plateauing or decline that can be defined. An overall process model is shown below and a series of eAgent support

will be connected to simplify and foster the often-painful process accepting the conditions faced and committing to and following through on treatment and care planning actions. A series of treatment and care action steps are tracked and managed in the next section.



When are you ready to do the Advanced Care Planning Process

Many Medicare beneficiaries lack either a short- or long-term care plan and often do not have a primary care doctor with geriatric experience. Fortunately, there is great research at NIH combined center from Institute of Aging and the National Institute of Neural Diseases and Strokes and has developed some recent significant recommendations that should be put forward as new efforts with NIH and CMS collaboration.

The eNavigator can work through a series of questions and guidance on issues such as:

- Personal Preferences and Level of Risk/Safety
- Review of Diseases of Aging and identify the “ounce of prevention steps” that can be made based on interventions that can keep patients away from expensive emergency rooms and hospital admissions and readmissions
- Identification of Neuropsychology of Cognitive Declining and Cognitive Fluctuations and triggers for stress and agitation based on “Behaviors that Challenge” and caregiver advice that can help managed the situation and built-in learning
- Establish a goal-oriented value chain strategy to prevent falls, infections, poor medical compliance and fostering new healthier behavior

The eNavigator can follow a pattern shown below:

Life Care Coaching and Scenarios Interviewing with questions such as:

- Life Care Needs and Fears: Aging Assessment
- Who will help me?
- Personal Health Risks
- What is the financial needs
- Conduct a "Ounce of Prevention" assessment and series of small interventions
- Conduct a Neurological Assessment and establish prevention of Brain Health including aging health literacy
- **Hire help.** If your budget allows, arrange for grocery delivery, a cleaning service or lawn care. Connect with professional Home Health Agencies and local office of Aging social workers,
- **Take your time.** When possible, do 1 chore each day, rather than having a marathon cleaning each week. You also can break work into smaller tasks. For example, take your holiday decorations out of the closet one day. Then set them out over the course of the week rather than all at once.
- **Declutter.** Simplify your life and give away things you no longer need. It gives you less to maintain.
- **Make cleaning easier.** Use cleaning tools, like dusters, with long handles so you can clean low or high without bending or reaching. Put knickknacks in a cabinet behind glass doors, so they don't have to be dusted as often.
- **Get organized.** Make a list of tasks that need to be done seasonally. This not only helps you remember what needs to be done but can save you time and money if you hire help.

"Once of Prevention" Assessment and Integrated Small Steps in Daily Schedule

Ignoring your health and hoping that walking, pickleball and beer will result in improved health is far too common among the aging baby-boom population. But the rush to the emergency room and the acute care hospitals for preventable situations increase. There are many different concepts over the years and some discussion about preventable hospitalizations and ideas that have not been deployed widely in preventive medicine. Assessment and daily small steps can be done to address issues in the following areas:

- Falls and Mobility
- Eating and Good Nutrition
- Avoid Bad Habits – Smoking and Heavy Alcohol Use
- Exercises including walking and weights
- Preventive Brain Health and Brain Exercises
- Vaccines and seasonal risks
- Emotional and Social Support
- Spiritual Healthy Behaviors

The "Once of Prevention" steps must be an acceptance by aging and those with disabilities that they have to reduce their risk and improve their safety. There has been extensive discussion by primary care physicians during the annual medicare appointment of conducting a risk assessment, but the effectiveness is questionable, High risk patients that receive care coordinated health and care services can build in "Once of Prevention" set of steps with the target improving quality of life and reducing hospitalization. Money can be saved.

Prepare an agitation and stress management large language model that can translate the "behavior that challenges" and general recommendations into an caregiver agitation management agentic ai-based advisor that support the caregivers.

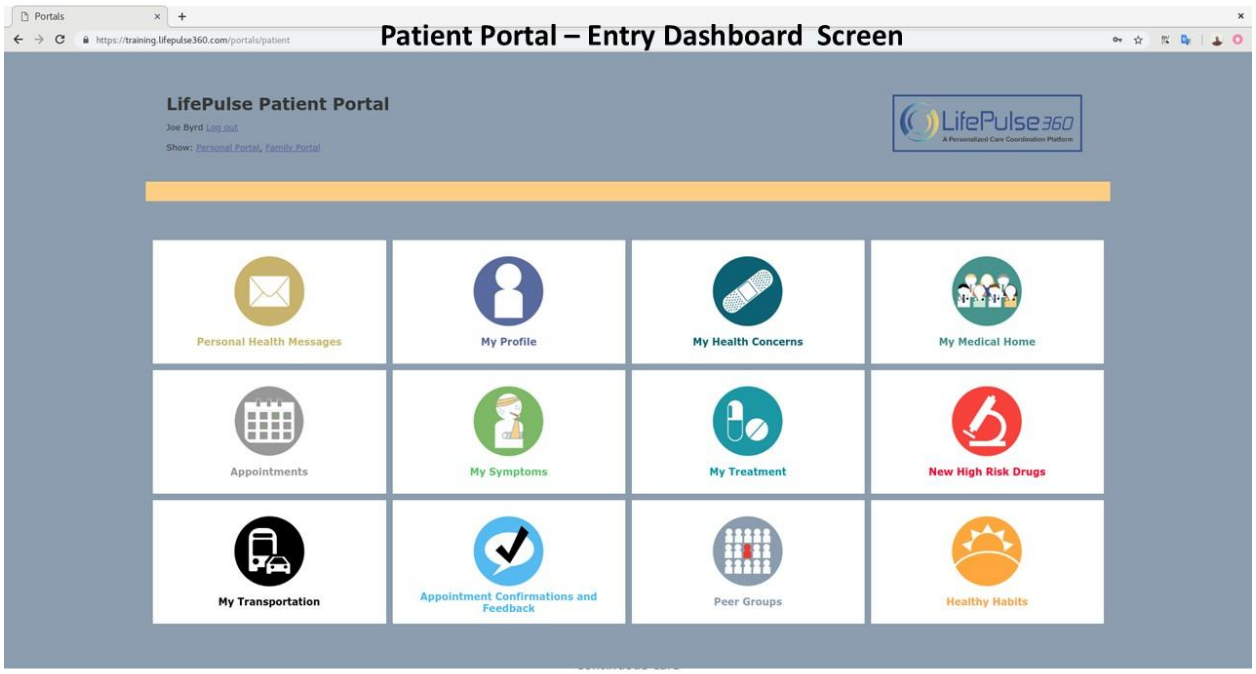
These interfaces for caregivers must take a simple approach such as discussed below.

Blue-Button-Like Interface for the Many Concerns with Users Agents to Empower User Management and Alignment of Personal Health Information and Progress Tracking with Advisory Agents and Recommenders

User experience on portals is difficult to use for beneficiaries, patients and caregivers. Blue button needs to be expanded to address the set of personal concerns of the patient with an panel of buttons shown below with each button linked to an underlying user agent backed with eAgents linked to an underlying agentic ai framework. Instead of the patient-caregiver relying on a doctor or expert they see every few months or twice a year to have your data. The patient caregiver owns the data and can share and align the data at each visit. Today, there is medication alignment but they is not supported with a automated process.

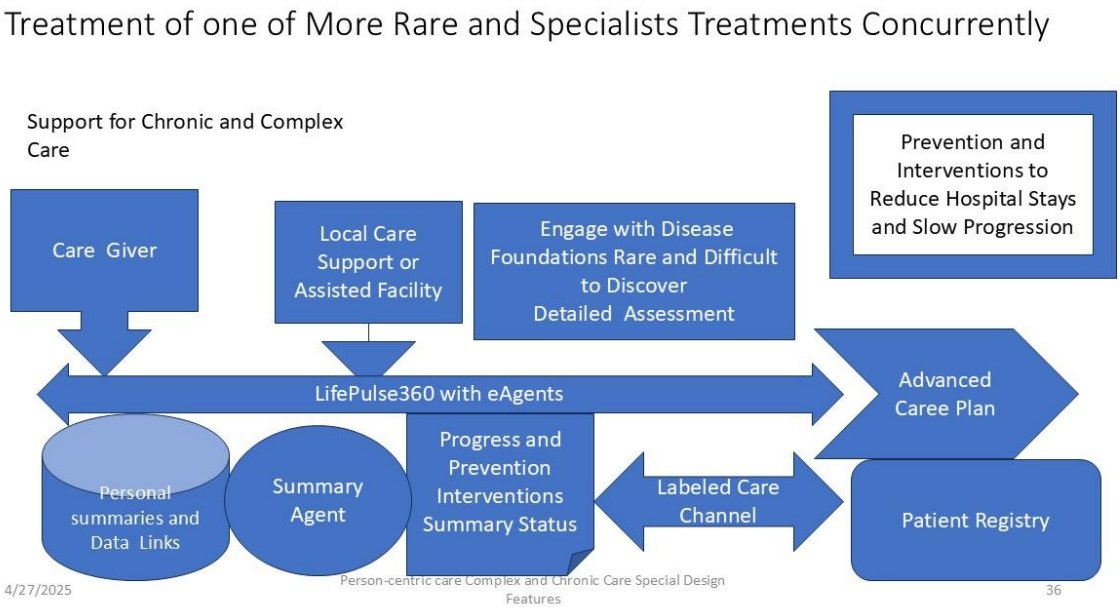
Providers give out their prescriptions and use their best techniques for fostering behavioral change with "motivational interviewing" and recommendations that are supposed to be very understandable. Of course, your told to ask any questions on the portal. This is after getting a scarry message on a diagnosis when the memory isn't working well now. The recent series of digital health or medicine articles describe the problems

User experience must be very simple using blue-button type approach but supported by a specific agent underneath. An example with categories of topics- using icons and visualization. This display can be adjusted by the user or caregivers. This interface can be displayed on all smart phones and computer interfaces.



Navigators Can Be Used in other areas such as Disease Discovery Below

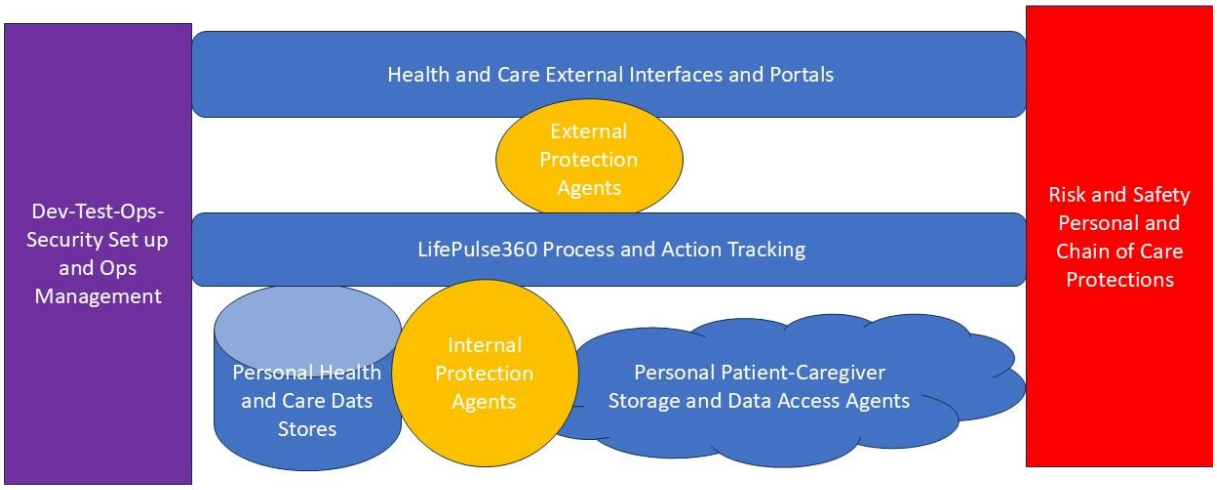
The care coordinators and patients need help to navigate through not only selecting insurance choices but safe places to live in a variety of housing conditions and to manage the rare and chronic conditions. This coordination is very complicated and can be shown in process diagrams like shown below and built with empowerment tools and stored in an open-source shared environment that fosters the people sharing problems and solutions in a systematic manner



Personalization with Links, Trust and Consent Management

There are both rewards and risks to having an integrated collection of both structure and unstructured data about a patient and caregivers. With fraud being rampant and fear of too much unauthorized and untrusted collection of personal data from multiple government sources – who can you trust and how can trust be managed. There have been extensive discussions and early progress on AI and risk and trust. Designing and building in consent, trust and active monitoring and management can follow the diagrams below and can use a series of consent and protection agents integrated into levels of protection.

Multi-Level Patient-Caregiver Operational Model for LifePulse360 with Outside and Inside Protections



Personalized Metrics and Population Metrics with Built-in Analytics

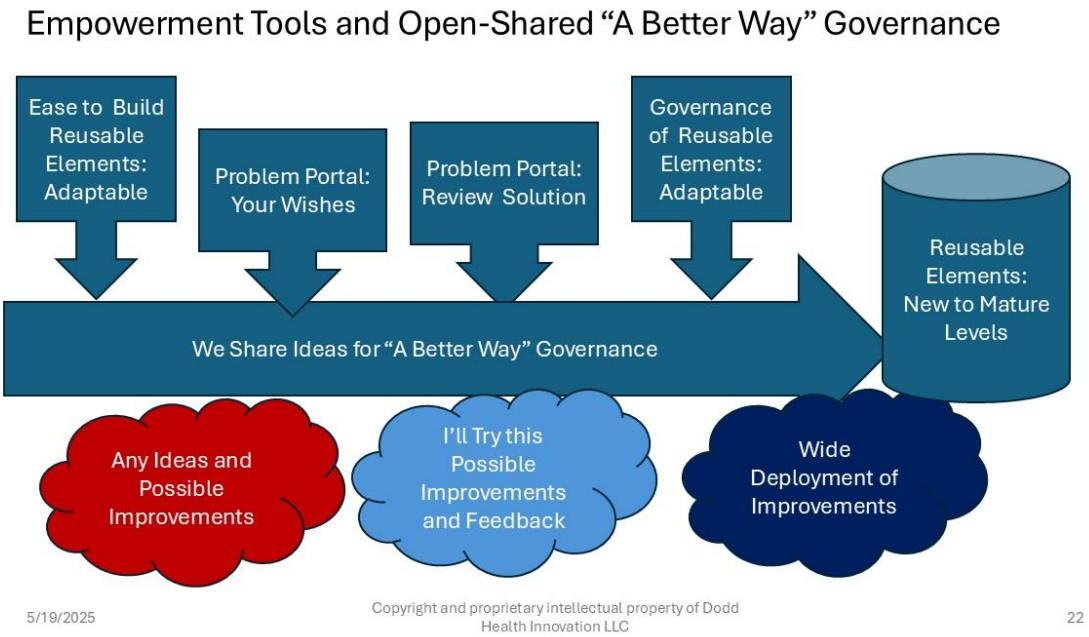
A new guided beneficiary-patient-caregiver environment such as LifePulse360 can meet both personal goals and expectations for insurance, treatment and care management. Measures can both reflect improved satisfactions but also reductions in expenses and improvement in life satisfiers. Personal metrics can create a dashboard, and volunteers can share their results in anonymized form with population-specific registries such as those by the National Organization of Rare Diseases (NORD) or state or federal metrics collection systems such as those managed by CMS Quality Management Information Systems. It is very important that analytics be built in and collection done automatically. A Collection-Analytics Agent is being designed with periodic transfer to registries/data lakes that have been agreed to,

There are of number of metrics but they need to be goal directed and personalized.

The HEDIS quality metrics for effective care could be personalized and linked to the patients advanced care plan along with their past utilizations and the goals for the next years' treatment and care planning. The HEDIS metrics could relate to the Medicare Advantage plans range of benefits and reason the patient-caregiver choose to select one plan versus another and the feedback after the next year.

Empowering Tools to Focus on Your Needs with Visibility and Adaptability

The eAgents fit within the current environment and are part of your personal portal, progress tracking dashboard and connects to the providers and hospital systems using open standards. The eAgents can be adapted and adjusted to personal needs and the communities and situations faced by the patient-caregivers. The complexity and confusion in medical and care management is hidden and managed by eAgents.



Innovation can come from a problem-solver (eg. John Dodd) that is faced with personal-family challenges and "said there is a better way". We feel that there are many such senior citizens with challenges and their ideas and solutions can be addressed with tools that create reusable and sharable interventions. LifePulse360 can create an environment and provide innovative governance for empowerment. It can be used by many disease foundation organizations and many advocates. It can foster grassroots innovation and will need the government to be an enabler with key foundational elements that build on TEFCA and USCDI and related standards and AI frameworks. For ongoing purposes, the initial work of ASTP/ONC and CMS can be turned over to Open NIEM and a health patient-first service committee that can use its established governance process to be part of a long-term governance process lead by the patient-advocate and disease foundations non-profits. The open-shared information exchanges along with the new generation of data and afentic AI services offer a new way to deliver services.

Reducing the Burden of Compliance with New Program Integrity Checks

The most recent Budget Reconciliation Bill has many program integrity checks for all beneficiaries of Medicare, Medicaid and ACA-Based Insurance Exchanges to address income from IRS, work requirements from labor and it is unclear how community engagement and compliancy rules can be defined. These multiple-program integrity checks need a common exchange approach such as we have proposed. The approach here must not only improve the existing systems but provide an alternative approach is needed or the new bill will increase and not decrease burdens to Beneficiaries. A new way forward must give the Beneficiary more choices with trusted and clear information about the dynamics and inconsistently of Medicare Advantage Programs. The streamline Medicaid Enrollment can leverage many of the technologies and concepts described here. The new focus on Pharmacy Benefits Management can also leverage the approach described. We can do better with the right technology.

As the LifePulse360 vision for improving the planning and engagement of patient and caregivers the underlying foundational technology agentic AI and medical large language models are evolving. The whole

area is evolving very quickly and the LifePulse360/Snowflake will track and welcome the opportunity to engage with ASTP/ONC and CMS.

Foundation of AI and Cloud-Based Technology with Snowflake Ready Now

SNOWFLAKE: AI DATA CLOUD FOR HEALTHCARE & LIFE SCIENCES

Snowflake provides simplified data architectures and deploy AI to make the most of healthcare analytics. Improve health outcomes, go to market faster, optimize supply chains and build patient and member 360s.

UNLOCK THE NEXT GENERATION OF HEALTH INNOVATION

Snowflakes mission is to break down data silos and enable secure data collaboration across the healthcare and life sciences industry, and the essential partners and technologies that support them.

Using Snowflake as a new empowerment platform will tap into an open, interoperable ecosystem that helps providers, payers and life sciences organizations overcome complex data architectures to improve health outcomes and reduce costs while maintaining the highest standards of data governance and privacy.

PATIENT AND MEMBER 360 FITS WITHIN THE LIFEPULSE360 CONCEPTS

Personalize care with a holistic view of patients and members that combines multiple data sources.

Snowflake enables easy access to unstructured, semi-structured and structured data from a single platform with near-infinite scale, which helps organizations generate holistic, real-time patient and member views across clinical, claims, consumer and socio-economic data silos.

Deliver Personalized Care

Implement patient and member personalization across channels while maintaining trust and privacy controls with one all-encompassing data repository to identify personalized patterns, trends, and correlations at scale.

Transform Patient and Member Experiences

Enable AI-powered virtual assistants and chatbots with advanced AI / ML workloads to personalize patient and member experiences and reduce administrative burden on care teams.

Snowflake Platform has a growing group of partners with the count changing everyday:

- 30 Data Health and Life Science,
- 6 technology partners
- 5 providers and payers
- 5 Pharmaceutical and Medical device

- 6 Digital Health and Tech

Month by month the family of health and life science partners is rapidly growing. It seems to be natural next step to move toward patient-caregivers is one of the next natural steps for Snowflake and interesting in looking forward to sharing our vision with ASTP/ONC and CMS.

Patient and Caregiver LLM-base Services

A number of the patient-caregiver concepts prevented for Insurance planning, care planning and Rare Disease Discovery, and management need to use Large Language Model-based services

Snowflake Cortex provides a fully managed service that enables access to industry-leading large language models (LLMs) is now generally available. Each problem-solving team can use these LLMs in specific states and connect with regional and international experts through a rare disorder network. Our sets of disease focused teams can build empowered tools to work directly via LLM Functions on Cortex. It enables problem-solving teams to bring generative AI securely with personal data and knowledge of their disease situation and using the patients and specific disease group governed data. Your team can focus on building AI applications, while we handle model optimization and GPU infrastructure to deliver cost-effective performance.

Here is the full set of updates released today as part of our mission to provide efficient, user-friendly and trusted generative AI:

- **More high-performing LLMs:** In addition to high-performing LLMs from Mistral AI and Google, Cortex now also supports Snowflake Arctic, Llama 3 (both 8B and 70B) and Reka-Core LLMs. All of these foundation models are available via the single, easy-to-use COMPLETE function - now generally available

- **Efficient RAG and semantic search using Arctic embed:** Embed and vector functions are now in public preview and include support for Arctic embed, the world's best practical text-embedding model for retrieval.

- **Enhanced AI Safety with Llama Guard:** As part of a collaboration with Meta, Llama Guard, an LLM-based input-output safeguard model, comes natively integrated with Snowflake Arctic, and will soon be available for use with other models in Cortex LLM Functions.

Health Innovation can be spurred by tracking and leveraging the many agentic ai and LLM advancements and turning them in specific focuses such as patient and caregiving needs. This effort

captures such an effort and it welcomes the opportunity to work with others in and outside of government.

Next Steps: Discussion and building a Collaborative Partnership for Patient-Caregiver Delivery Redesign

A more detailed discussion of Partnering and Implementation Approaches will evolve into an eBook and demo's of the existing features of Empowering Beneficiaries with LifePulse360 and a new generation of AI tools can turn the attention to patients and caregivers.

The question is there a market that is ready to these ideas. Are we ready to support innovations. Can the government foster change and empower the communities to build process and knowledge-driven tools, is a follow-up "21st Care Implementation Act" needed. Can we take the next steps towards personalized interoperability and knowledge sharing. Can we close the research expertise to practice gaps. We hope our ideas can trigger further discussion and be the next step to health innovation transformation steps.

Please contact John Dodd at 410-598-7266 or john.dodd@bdc-healthit.com.